SELF-EVOLUTION SHOULD NOT DEPEND ON HOW SKILLS ARE SPLIT: AN EXACT CERTIFICATE FOR SKILL-TAXONOMY REPRESENTATION INVARIANCE

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ABSTRACT

Self-evolving language-model agents increasingly treat skill identities as control objects: a sampled skill can determine the next training task, a fresh ID can receive retrieval priority, and feedback can be written back to the selected package. We ask whether such control should change when semantic capabilities are merely split or cloned across package identities. We formulate *Skill-Taxonomy Representation Invariance* (STRI). For exact refinements, we prove that invariance is equivalent to factoring aggregate semantic-class mass through the quotient; any strictly positive identity-local score followed by normalization is therefore clone-sensitive. Released Skill-SP instantiates this failure mode at initialization: 15 packages receive probability $1/15$ each, double-supported Level-1 rows receive twice the additive eligibility opportunity of singleton rows, and a counterfactual same-state sampler-input clone changes its prompt-mixture by TV $= 0.0583$; quotient-conserved mass restores it exactly. For partial overlap, define $R^*(A; q) = \min\{t : q \leq Aw \leq tq,\ w, t \geq 0\}$ for support matrix A and representation-independent target $q > 0$. It equals one iff q lies in the package support cone. Skill-SP Level-1 has neutral $R^* = 2$ on full, calibration, and tool-disjoint heldout matrices. Uniform routing already attains this exact worst-case optimum, while inverse-support and NNLS target fitting worsen worst-case distortion to 98.3 and 5.51 despite the latter lowering average dispersion; calibration-fitted exact weights transfer to heldout at $R = 2$ without refitting. Level-3, the 127/128-overlap logical domain, and a 75-query expert SkillRouter relevance-graph analogue are equalizable. Across these regimes, target non-realizability—not overlap prevalence or poor scalar tuning—is the audited discriminator. In one archived AutoSkill P19 substrate, splitting one skill into four exact-semantic IDs deterministically crowds out a post-checkout skill and removes a mechanically defined destructive behavior (6/6 original versus 0/6 split), while ID-placebo and quotient controls restore it (3/3 each; Fisher $p = 0.00108$). Under the split representation, adding that post-checkout skill back restores the behavior in 3/3 fresh runs while a matched cleanup add-back yields 0/3 (exact one-sided Fisher $= 1/20$), isolating the mediator within this substrate. Thus representation can alter executed behavior on this bounded substrate without semantic change. A semantic-first construction marks the action-basis escape when package-only control cannot realize q; it is constructive, not a validated repair.

1 INTRODUCTION

Many self-evolving agents adapt through external state—memories, skills, workflows, tools, prompts, retrieval policies, or harness code—rather than only through foundation-model weights. Skills are especially attractive because a skill can package reusable procedure, metadata, examples, applicability information, and verification logic. Recent systems make the skill identity itself a control variable: Skill-SP (Huang et al., 2026) samples a skill before generating a training task, while SkillRL (Xia et al., 2026) gives dynamically created skill IDs priority in a fixed retrieval budget.

This architecture makes the *representation* of the skill library consequential. Consider two taxonomies that expose the same executable semantic capability. One stores a primitive once; another stores an exact clone under a fresh ID. One represents generic and specific capabilities separately; another groups unchanged primitives behind a macro package. If no new primitive behavior or support region is added, should the future curriculum, retrieval exposure, or credit allocation change solely because package bookkeeping changed?

Prior work establishes that skill availability, representation, retrieval, composition, and exposure matter (Xu & Wu, 2026; Jiang et al., 2026; Gao, 2026; Singh & Joachims, 2018; Biega et al., 2018); Sec. 2 separates our support-fixed estimand from those objects. The challenge is that overlap and skill count themselves change under refactoring. We therefore work asset-first: freeze independent support truth, quotient exact-clone nuisance, overgrant the strongest same-information package baseline, and ask whether the remaining support geometry can realize a representation-independent target.

Released Skill-SP. On API-Bank Level-1, 314 released tasks are covered by at least one released validator and 183 have multiple memberships. The release's name+description Jaccard duplicate filter at threshold 0.33 misses all five observed specific–generic support-overlap pairs. Minimum-cardinality whole-package pruning can reduce the active library from 10 to 7 while preserving exact support, but 71 overlapping tasks remain. The residual is therefore not merely an exact-clone artifact.

Released SkillRL. In released template-mode retrieval, dynamic general skills are included before the remaining fixed `top_k=6` budget is filled with static skills. Adding one exact-content fresh-ID copy at a time admits all 12 tested clones. Across 223 released ALFWorld task descriptions, 11/12 clone targets change the unique semantic retrieval set and 5/12 reduce the number of unique general-skill contents returned.

These systems establish representation sensitivity, but not its boundary. The logical domain provides a direct counterexample to overlap-as-sufficient-explanation: 127/128 tasks overlap while the neutral target remains exactly realizable. Thus overlap count alone cannot determine the STRI residual.

Contributions.

1. **A support-fixed systems invariant.** We ask whether a skill-indexed self-evolution controller changes under package refactorings that preserve semantic support. The novelty claim is this invariant and audit object, not skill representation or exposure allocation in general.
2. **An exact certificate.** Quotient factorization characterizes exact-refinement invariance and $R^*(A;q)$ decides package-first target realizability; semantic-first selection marks the action-basis escape. We reuse classical convex feasibility/duality rather than claim a new LP or cone theorem.
3. **Released-system evidence with explicit boundaries.** Skill-SP/SkillRL expose identity sensitivity; practical package-weight baselines and no-refit heldout transfer separate worst-case invariance from average fitting; positive/negative support geometries plus an external expert relevance analogue separate overlap from realizability; one frozen AutoSkill P19 intervention traces a bounded representation-to-behavior path. Structured support, target, witness, and retrieval-budget stresses quantify robustness without expanding to general utility or safety.

2 Relation to Existing Work

Controlled skill representation changes a different variable. SkillsBench varies skill availability and presentation granularity (Xu & Wu, 2026); Demystifying Agent Skills isolates representation, annotation, retrieval difficulty, and cross-framework robustness (Jiang et al., 2026); SkillRouter evaluates large-pool retrieval over 75 expert-verified SkillsBench-derived queries (Zheng et al., 2026); composition work maps task decompositions to multiple skills (Gao, 2026). They establish that presentation and retrieval matter, but do not hold executable semantic support fixed while perturbing only package bookkeeping. We use SkillRouter's released relevance graph only as an external negative analogue, never as executable support truth.

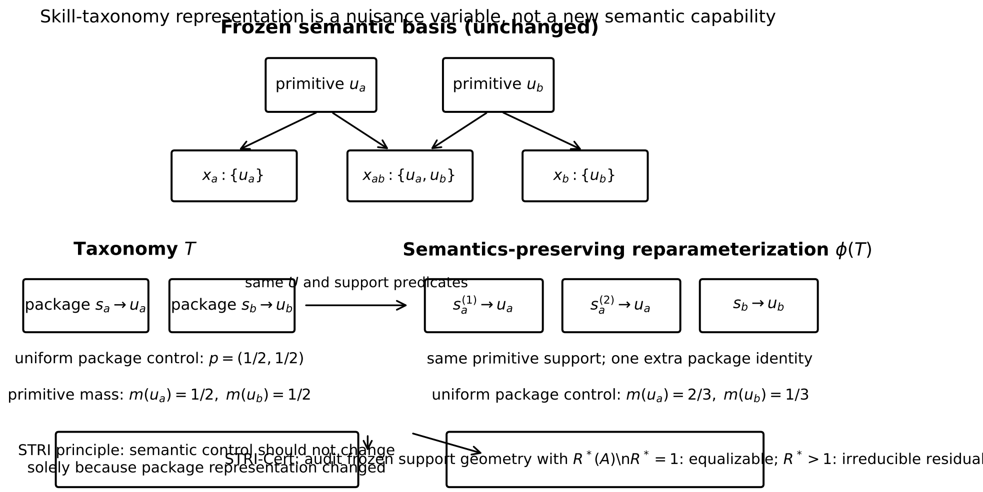

Figure 1: STRI separates semantic capability from package representation. The conceptual identity-split example leaves the primitive basis and support predicates unchanged, yet a package-indexed controller can change primitive mass solely because one primitive has two package identities. Exact clones motivate the nuisance channel; the decisive released residual in Sec. 6 survives clone reasoning and is characterized by non-equalizable partial-overlap support geometry.

Self-evolving controllers are the closest substrate. Skill-SP (Huang et al., 2026) is closest: it samples a skill before task generation and releases independent validators. STRI then overgrants its package controller, allowing any nonnegative mass under the full frozen support matrix. SkillRL (Xia et al., 2026) supplies an identity-sensitive fixed-budget retrieval phenotype but no independent A; AutoSkill (Yang et al., 2026) supplies one bounded behavioral seam. Work on redundancy, failures, routing, and skill/harness co-evolution motivates the setting but not this support-fixed audit (Zhu et al., 2026; Liu et al., 2026; Mao et al., 2026; Dong et al., 2026; Chen et al., 2026; Wang et al., 2026).

Optimization boundary. Fair-exposure methods allocate attention (Singh & Joachims, 2018; Biega et al., 2018); our covering/cone/duality machinery is classical (Lovász, 1975; Sinkhorn & Knopp, 1967; Radunović & Le Boudec, 2007; Beck & Fiala, 1981; Boyd & Vandenberghe, 2004). We claim the systems invariant, information boundary, and audit mapping—not a new LP, cone theorem, or fairness objective.

3 Method: From Package Identity to Semantic Exposure

3.1 Protocol at a glance

Design intuition. Package j contributes support column $A_{:j}$, so package mass w induces semantic exposure Aw. In Fig. 1, an exact clone is therefore a duplicated support column: it may perturb identity-indexed package mass, but should not create a new semantic degree of freedom. STRI-Cert separates two questions: exact identity refinements should vanish after quotienting clones; afterward, a target q is package-realizable exactly when it lies in the nonnegative cone of the remaining columns.

The five-step order isolates those questions: **freeze support/target provenance** $\rightarrow$ **quotient exact clones** $\rightarrow$ **overgrant arbitrary** $w \geq 0$ **with full frozen** A $\rightarrow$ **solve primal/dual** ($R^* = 1$ constructive weighting; $R^* > 1$ tight residual plus witness) $\rightarrow$ **separate diagnosis from repair**. Semantic-first control can escape cone infeasibility only by changing the action basis, so it is a constructive boundary witness rather than validated downstream repair.

3.2 FROZEN SUPPORT MATRIX

Let $X = \{x_1, \ldots, x_n\}$ be a finite set of semantic contexts or tasks with independently defined first-party support truth, and let $S = \{s_1, \ldots, s_m\}$ be the current skill packages. Define the binary incidence matrix

$$A_{ij} = \mathbf{1}\{s_j \text{ supports } x_i\}.$$

Support is frozen before STRI: released Skill-SP validators define the tool-call matrix, while released compiler specifications plus fixed CSP/contract checks define the logical matrix. Certificates cover only rows with at least one supported package; zero-support rows are reported separately rather than misclassified as representation distortion.

At a frozen pre-task state, package-only control assigns $w \in \mathbb{R}^m_{\geq 0}$ and induces eligibility exposure $e = Aw$. Exposure is control opportunity, not task probability or utility. The target $q \in \mathbb{R}^n_{>0}$ is a *representation-independent semantic target*: it cannot change because packages are renamed, cloned, split, or regrouped. Without an externally justified preference we use neutral $q = \mathbf{1}$.

3.3 SAME-INFORMATION PRE-TASK CONTROLLERS REDUCE TO PACKAGE MASS

In released Skill-SP, skill selection occurs before task generation. At this decision point the action is one current package identity. For any fixed pre-task state z, every randomized same-information rule—including a bandit, MLP, or neural router—induces a categorical distribution over packages, hence one nonnegative mass vector $w(z)$. The strongest same-information baseline is therefore arbitrary nonnegative package mass with access to the complete frozen support matrix.

This is an intentional overgrant: the baseline may ignore the released scoring rule and choose any $w(z)$ using complete frozen A and the same pre-task information. If this class has $R^*(A; q) > 1$, deduplication, retuning, bandits, or neural package routers cannot remove the residual without new information or a different action basis. The reduction is pointwise in z; it neither collapses a longitudinal controller to one static vector nor permits conditioning on the future task.

3.4 EXACT-REFINEMENT STRI FACTORS THROUGH A QUOTIENT

Let $\mathcal{R}$ be a package representation and let $Q(\mathcal{R})$ collapse exact semantic clones—packages with identical semantics, support, and controller-relevant state—into classes. Let $G_{\mathcal{R}}$ sum a package-mass vector over these classes. For a fixed pre-context state z, define *exact-refinement STRI* by

$$G_{\mathcal{R}} p_{\mathcal{R}}(z) = G_{\mathcal{R}'} p_{\mathcal{R}'}(z) \quad \text{whenever } Q(\mathcal{R}) = Q(\mathcal{R}').$$

Proposition 1 (quotient-factorization characterization). A controller family satisfies exact-refinement STRI iff there exists a quotient-level map $\bar{p}$ such that $G_{\mathcal{R}} p_{\mathcal{R}}(z) = \bar{p}(Q(\mathcal{R}), z)$ for every representation and state.

Proof. The forward direction defines $\bar{p}$ using any representative of a quotient class; STRI makes this definition representative-independent. The reverse direction is immediate because equal quotients give equal $\bar{p}$. □

Corollary 1 (identity-local normalization is clone-sensitive). Suppose $p_j = g(h_j, z)/\sum_k g(h_k, z)$ with $g > 0$ and an exact clone receives the same local state h_j. Writing its score as $a > 0$ and all other scores as $B > 0$, the semantic-class mass changes from $a/(a + B)$ to $2a/(2a + B)$, an increase of $aB/((a + B)(2a + B)) > 0$. Repeated clones drive class mass toward one, so exact-refinement invariance requires class-aware conservation or another quotient-factorized rule.

Proposition 2 (semantic-first construction handles arbitrary overlap). Normalize a target ray $q > 0$ to $\pi = q/(\mathbf{1}^\top q)$. If every target-positive semantic unit i has at least one supporting package, choose any row-stochastic kernel K with $K_{ij} = 0$ whenever $A_{ij} = 0$, then sample $i \sim \pi$ before sampling implementation $j \sim K_i$. The semantic marginal is exactly π for arbitrary overlap, and any clone/split can redistribute K_i within the corresponding implementation class without changing that marginal.

Proof. The joint mass is $J_{ij} = \pi_i K_{ij}$, so $\sum_j J_{ij} = \pi_i \sum_j K_{ij} = \pi_i$; support-constrained row-stochastic K exists exactly when every positive-target row is covered. □ This changes the action interface: it requires representation-independent semantic intent and conditional implementation. It is a constructive action-basis witness for escaping package-first infeasibility, not a same-information baseline or validated downstream method.

4 Target Realizability and STRI-Cert

4.1 Target-conditioned finite-support criterion

Define the multiplicative distortion from the target ray q to the package support cone by

$$R^*(A; q) = \min_{w \geq 0,\, t \geq 0} t \quad \text{s.t.} \quad q \leq Aw \leq tq. \tag{1}$$

We write $R^*(A) \equiv R^*(A; \mathbf{1})$ for the neutral audit used in our released-system tables.

Theorem 1 (target realizability in the support cone). For any $q > 0$, $R^*(A; q) = 1$ if and only if $q \in \mathrm{cone}(A) = \{Aw : w \geq 0\}$. Otherwise $R^*(A; q) > 1$ is the tight minimum componentwise multiplicative distortion from q achievable by package-only mass.

Proof. The constraints force $t \geq 1$. At $t = 1$ they reduce exactly to $Aw = q$, which is equivalent to $q \in \mathrm{cone}(A)$. For any feasible w, the smallest admissible t is $\max_i (Aw)_i / q_i$ after scaling w so that $\min_i (Aw)_i / q_i = 1$; minimizing over w gives the tight distortion. □

This separates STRI from a uniformity assumption: $q = \mathbf{1}$ is the neutral target, not the definition of representation invariance. It also makes exact clone/split invariance immediate: duplicating a support column does not change $\mathrm{cone}(A)$, so it cannot change $R^*(A; q)$ for any fixed q. The cone-membership statement itself is classical convex feasibility; STRI contributes the system invariant that makes this feasibility question the relevant audit object for a self-evolving skill controller, not a new theorem about convex cones.

Proposition 3 (dual certificate). The dual of Eq. (1) is

$$D^*(A; q) = \max_{\alpha, \beta \geq 0} q^\top \alpha \quad \text{s.t.} \quad A^\top (\beta - \alpha) \geq 0, \quad q^\top \beta \leq 1. \tag{2}$$

Strong duality gives $D^*(A; q) = R^*(A; q)$. Thus every dual-feasible pair is a certified lower bound, while an optimum is complete even when no local witness template is available. On every audited neutral-target regime the numerical primal–dual gap is below 10^{-8}.

4.2 Human-readable factor-2 witness

Theorem 2 (global-singleton overlap lower bound). Suppose packages a, b each have a context supported by exactly that one package over the full frozen support universe, and another context is supported by both a and b, possibly with additional packages. If both singleton contexts retain positive exposure, then every nonnegative package mass has max/min exposure ratio at least two on the focal rows, and therefore $R^*(A) \geq 2$ for the full support matrix.

Proof. Let $r > 0$ be the minimum exposure among the two singleton contexts and the shared context. Since the first singleton support signature is exactly $\{a\}$, its exposure is $w_a \geq r$; likewise $w_b \geq r$. The shared context contains both a and b, and additional packages have nonnegative weight, so its exposure is at least $w_a + w_b \geq 2r$. Thus every w has focal max/min ratio at least two. For the same w, the full-matrix maximum is no smaller than the focal maximum and the full-matrix minimum is no larger than the focal minimum, so the full-matrix ratio is also at least two. Minimizing over w gives $R^*(A) \geq 2$. □

For the neutral target $q = \mathbf{1}$, Theorem 2 is also a sparse dual certificate: put $\alpha = 1$ on the two singleton rows, $\beta = 1$ on their shared row, and zero elsewhere. Dual feasibility follows from the support pattern and the objective is two. In released Skill-SP Level-1, the LP dual recovers such a sparse optimum and the primal attains $R^* = 2$.

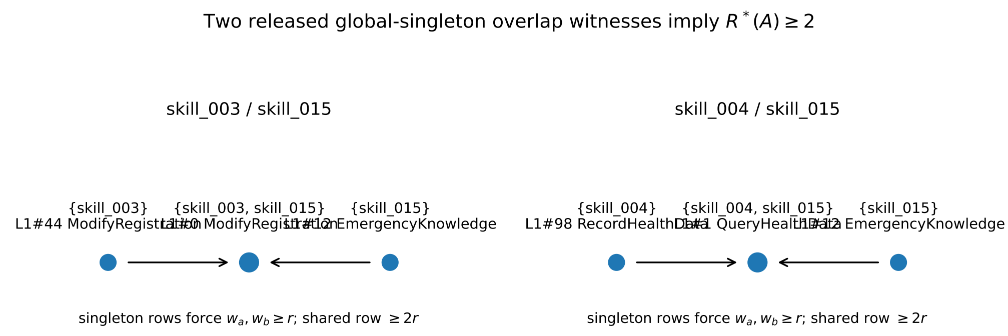

Figure 2: Two released global-singleton overlap witnesses. In each case, retaining positive exposure on the two singleton regions forces at least twice that minimum exposure on the shared region.

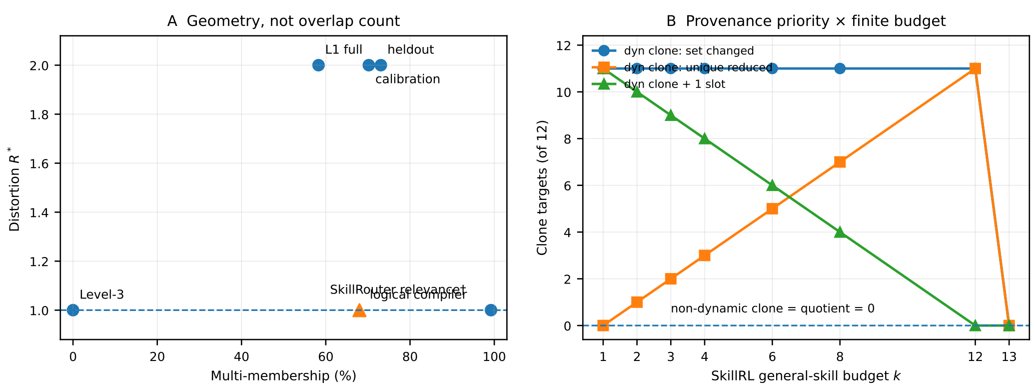

Figure 3: Boundary breadth. **A:** support geometry separates residual/equalizable regimes; SkillRouter† is an expert relevance analogue, not executable support. **B:** SkillRL exact clones change semantic retrieval only under fresh-identity priority plus finite budget; placebo/quotient controls stay at zero and capacity closes the semantic-set effect.

4.3 STRI-Cert and deployable variants

STRI-Cert is a diagnostic protocol, not a learned predictor or a new LP solver. Its output has three operational meanings: **realizable** ($R^* = 1$) means package-only retuning is sufficient *in principle* to hit the declared target; **residual** ($R^* > 1$) means no same-information package router can hit that target without changing information, assumptions, or the action basis; and **fail-closed** means support/target provenance or optimization is not valid enough to make either statement. The dual may expose a sparse obstruction, while the primal gives a constructive weighting in the realizable case. HiGHS solves both programs with paper-level tolerance 10^{-8}; audited primal–dual gaps are at most 10^{-8}. The certificate concerns feasibility, not downstream utility or a unique weight vector.

Two variants make deployment assumptions explicit. With independently justified graded support $0 \leq L \leq A \leq U$, the box-robust audit uses $Lw \geq q$ and $Uw \leq tq$; estimating valid bounds remains an identification problem. Adding $w_j \leq \rho \sum_k w_k$ audits package-mass concentration via $R^*_\rho(A;q) \geq R^*(A;q)$. Both remain linear programs.

5 Experimental Setup

Evidence, estimands, and controls. Predictions are direct: clone/split refactoring leaves the support cone and R^* unchanged; overlap alone need not imply residual; a factor-2 witness rules

Table 1: Two-stage audit across released systems. We test observable identity sensitivity first and run STRI-Cert only when independent support truth exists; absence of a support contract is not repaired with candidate-defined labels.

System	Identity perturbation	Independent A?	Audit role
Skill-SP	Pristine sampler gives $2\times$ eligibility; sampler-input clone changes class mass $1/15 \rightarrow 1/8$, while quotient conservation restores it exactly	Yes: released validators	Controller violation + exact clone repair + geometry certificate
SkillRL	Fresh-ID exact clones change semantic retrieval for 11/12 targets across $k = 1$–12; effect vanishes at $k = 13$	No explicit support contract	Budget-bound phenotype + placebo/quotient controls; no R^*

Table 2: Practical package-weight baselines. Entries are neutral-target worst-case exposure ratios (lower is better). "Heldout" freezes weights fitted on calibration tools; no heldout refit is allowed. Semantic-first changes the action basis and is shown only as an upper-bound construction.

Baseline	L1 full	Heldout	L3	Logical
Released uniform	2.00	2.00	1.00	8.00
Inverse support size	98.33	2.89	3.00	15.67
Inverse $\sqrt{\text{support}}$	10.87	2.38	1.73	11.02
NNLS target fit	5.51	3.98	1.00	1.00
Exact min-cover + uniform	2.00	2.00	1.00	1.00
Max–min fair	2.00	2.00	1.00	1.00
Exact R^*	**2.00**	**2.00**	**1.00**	**1.00**
Semantic-first†	1.00	1.00	1.00	1.00

out all same-information package weights for neutral q. Skill-SP tests positive/transfer regimes; Level-3/logical compiler test negative boundaries.

Six evidence roles. Structural audits use neutral $R^*(A)$ with primal/dual tolerance 10^{-8}. E1 is the released-system result; E2 practical package-weight/quotient baselines with calibration→heldout frozen-weight transfer; E3 negative regimes, the expert-relevance analogue, SkillRL budget/placebo/quotient sweeps, and AutoSkill's bounded P19 intervention; E4 support/target sensitivity; E5 witness redundancy and the P19 mediator trace; E6 conditional solver cost. A separate SkillsBench qualification rejects task-local availability as exact support (79/87 rows disagree with `required_skills`); it contributes no R^* result. New breadth analyses use frozen public artifacts only—zero new annotations, model calls, or GPU runs.

6 Experiments and Results

6.1 Skill-SP tool-call support and released-controller consequence

API-Bank Level-1 has 314 covered rows, 183 with multiple memberships, and neutral $R^* = 2$. The released pre-task sampler initializes all 15 tool-call packages at probability $1/15$, so a double-supported row receives $2/15$ eligibility versus $1/15$ for a singleton. The source paper reports a -1.9 point accuracy change when dynamic routing is replaced by uniform routing (Huang et al., 2026); we use this only to establish that allocation is operational, not to attribute utility to STRI.

For a same-state sampler-input reparameterization, we copy one active package with identical content, support, statistics, and sampler state under a fresh ID and rerun the author's weighting function. This is not a claim that the induction path would admit a literal duplicate. The cloned semantic class moves from $1/15$ to $2/16 = 1/8$, yielding prompt-mixture TV $= 7/120 = 0.0583$ because package ID is absent from the questioner message. Quotient-conserved mass instead splits the original $1/15$ class mass and restores prompt TV exactly to zero. These are input-mixture statements, not generated-task or utility claims.

Table 3: Neutral-target boundary across first-party support regimes and one external expert relevance analogue. High multi-membership is neither necessary nor sufficient; †SkillRouter labels retrieval acceptability, not executable support.

Regime	Covered	Multi	Overlap	R^*	Certificate
API-Bank Level-1 full	314	183	58.3%	2.0	Residual
Level-1 calibration tools	47	33	70.2%	2.0	Residual
Level-1 tool-disjoint heldout	52	38	73.1%	2.0	Residual
API-Bank Level-3	34	0	0.0%	1.0	Equalizable
Logical compiler validation	128	127	99.2%	1.0	Equalizable
SkillRouter core relevance†	75	51	68.0%	1.0	Equalizable analogue

A frozen tool-disjoint test gives the same structural answer: calibration has 47 covered rows/33 multi-memberships and $R^* = 2$; heldout has 52/38 and $R^* = 2$. More stringently, weights fitted only on calibration are frozen before heldout evaluation: uniform, exact min-cover, max–min, and exact-R^* weights retain ratio 2, while inverse-support and NNLS transfer at 2.89 and 3.98. Thus heldout residual does not require refitting to heldout rows. The released sparse witnesses match the exact LP.

6.2 Level-3: disjoint-support negative control

Level-3 provides a natural released negative regime. Thirty-four rows are covered, every covered row has exactly one skill membership, and $R^* = 1$. Equal weights on the active packages equalize every covered task. STRI-Cert therefore does not mechanically label every released taxonomy problematic.

6.3 Logical compiler domain: high overlap but no residual

The first-party logical release gives a stronger negative control: 128/128 programmatically generated units pass the author's compiler/uniqueness contracts and are cross-checked against all eight released support specifications, with no language model. Although 127/128 units have multiple memberships across 38 support patterns, $R^* = 1$ because `skill_008` supports every unit. This equalizer is concentrated: the max-share audit gives $R^*_{0.75} = 4/3$, $R^*_{0.5} = 2$, and $R^*_{0.25} = 4$. High overlap is therefore not sufficient; realizability depends on support geometry and the declared feasible package-mass set.

6.4 External relevance analogue and SkillRL budget boundary

SkillRouter releases expert task–skill relevance labels for 75 scored queries (24 single, 51 multi) (Zheng et al., 2026). Treating these labels strictly as a *retrieval-relevance analogue*, not executable support, uniform package mass yields a $7\times$ exposure spread, yet the core and all-gold graphs have $R^* = 1$; including score-1 degraded relevant variants raises uniform spread to $21\times$ while remaining equalizable. This independent negative graph reinforces that multi-membership and naive exposure spread do not imply cone non-realizability.

SkillRL supplies a complementary identity-sensitive retrieval seam. Across 223 ALFWorld descriptions and all 12 exact-clone targets, fresh-`dyn_` priority changes semantic retrieval for 11/12 targets for every $k \in \{1, 2, 3, 4, 6, 8, 12\}$; unique-content crowd-out grows from 0 to 11 targets. A same-content non-dynamic-ID duplicate and exact-semantic quotient cause zero semantic-set changes at every tested budget. At $k = 13$, where all 12 static contents plus the clone fit, the semantic-set effect vanishes; one extra slot at $k = 12$ also restores it. This is a provenance-priority$\times$finite-budget phenotype, not an R^* or utility claim; the separate qualified endpoint bridge is not a population-level no-effect theorem.

6.5 AutoSkill: dynamic behavioral propagation

AutoSkill (Yang et al., 2026) exposes a first-party `sdk.search` seam. In one frozen five-skill P19 library at `top_k=5`, a four-way exact-semantic split crowds out a post-checkout skill; ID renaming

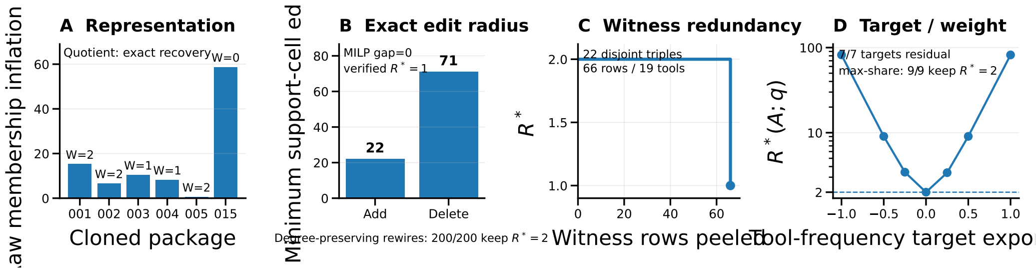

Figure 4: Structural robustness. **A:** clone perturbation and quotient recovery. **B:** exact edit radii are 22 additions/71 deletions; 200/200 degree-preserving rewirings retain $R^* = 2$. **C:** the residual survives 22 disjoint three-row witnesses spanning 19 tools. **D:** 7/7 frozen target rays remain residual and max-share $\rho \in [1/6, 1]$ leaves $R^* = 2$.

preserves, and exact-semantic quotienting restores, the original semantic set. The mechanical post-checkout behavior occurs in 6/6 original, 0/6 split, 3/3 placebo, and 3/3 quotient runs (one-sided Fisher $p = 0.00108$). Under the split, adding the crowded-out skill back restores behavior in 3/3 fresh runs versus 0/3 for a matched cleanup add-back ($p = 1/20$). This isolates one mediator on one substrate, not AutoSkill utility, safety, or prevalence.

6.6 Matched baselines, representation ablations, and failure analysis

Table 2 compares concrete package-weight controllers under the same frozen support. On Level-1, released uniform routing already attains the exact worst-case optimum 2, so the residual is not poor scalar tuning. Inverse-support heuristics worsen the full ratio to 98.33 (10.87 with square-root tempering). NNLS lowers exposure CV from 0.312 to 0.215 yet worsens worst-case ratio to 5.51, showing that average fit and STRI's worst-case objective are distinct. Exact min-cover, max–min, Chebyshev fitting, and R^* cannot beat 2. Minimum exact-coverage pruning still removes 61.2% of overlap without absorbing the residual; exact cloning/splitting leaves the support cone unchanged.

E4–E5: structural robustness. Across 49 tools, all 4 closed-form-witness subsets are residual while all 10 overlap-without-witness subsets are equalizable. Level-1 remains residual under all 1,387 one-cell additions, 366 valid deletions, 184 tool-block$\times$package biases, 49/56 signature relabels, and 200 degree-preserving bipartite rewirings; the logical analogue breaks under 127/595 valid deletions. Exact neutral-target MILPs require at least 22 added cells or 71 deletions to reach $R^* = 1$; 7/7 frozen target rays and all tested max-share constraints remain residual. Peeling yields 22 pairwise-disjoint three-row witnesses spanning 66 rows/19 tools (21.0% of Level-1) before the remainder reaches $R^* = 1$. Package-wide support overstatement remains a boundary (3/9 variants residual). These are finite-snapshot sensitivity/redundancy controls, not randomization significance or witness uniqueness.

E6: conditional solver cost. Dense SciPy/HiGHS grows from 0.00234 s median at 128×8 to 0.685 s at $16{,}384 \times 96$ (largest max 0.765 s; 32,768 inequalities; 24.25 MiB), a host-load-dependent profile with no asymptotic or solver-novelty claim.

7 Discussion, Limitations, and Takeaway

Skill-SP/SkillRL establish identity sensitivity with tuning/budget controls; frozen support regimes show geometry—not overlap count or average fit—separates residual from equalizable cases, and SkillRouter supplies an external relevance-graph negative analogue. AutoSkill remains one bounded behavior existence proof. STRI-Cert is conditional on A, q: $R^* = 1$ keeps package retuning viable, $R^* > 1$ rules out that same-information class, and invalid provenance redirects effort to identification. The AutoSkill path and SkillRL phenotype neither supports population utility, safety, or regret; semantic-first remains constructive, not a validated repair.

AI USE STATEMENT

In this work, generative AI tools assisted with conceptual framing and hypothesis refinement; formulation and critique of mathematical claims and proof drafts; experiment and falsifier design; implementation and review of analysis, experiment-control, and manuscript-quality-assurance code; interpretation of intermediate results; primary-source literature triage; and preparation or editing of manuscript text, figures, tables, and references. Generative models were also used as the task proposer in a preregistered Qwen3-8B dynamic pilot; that pilot failed its frozen author-native qualification gate, remains documented in the supplement, and is excluded from scientific claims about dynamic STRI. Load-bearing numerical claims in the submitted narrow paper are bound to released first-party artifacts or deterministic/programmatic analysis outputs. The support certificate, theorem edge cases, transition logic, and paper-level numerical/citation constraints were exercised by executable tests; related-work boundaries and bibliographic metadata were checked against primary or official sources. AI-assisted outputs were not treated as independent scientific authority. We take responsibility for the final content of this work, including text, claims, code, and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The main paper states the exact-refinement quotient characterization, semantic-first design construction, target-conditioned primal/dual programs, closed-form witness, box-robust extension, and constrained-controller audit used for the load-bearing claims. The Skill-SP controller audit is bound to author commit `bb693c89fee66e1f824d6a777759a49b7a295a83`, released `library.py` SHA-256 `8441de35...e2756e0`, question-generation SHA-256 `1b4cb67f...82d739b`, initial package-bundle SHA-256 `d4c4350a...68b6e7`, and tool-call membership SHA-256 `b73d8637...6f2e17d4`. The supplement packages our controller-audit/certificate code, tests, frozen receipts and matrices, but intentionally does not redistribute the third-party Skill-SP repository; its README gives the exact author commit required for end-to-end rerun. LPs use SciPy HiGHS; certificate classification uses tolerance 10^{-8}. The offline completion analysis is implemented in `asset_first_stri_offline_completion_analysis_20260824.py` and records the frozen Level-1 structured-misspecification families plus deterministic synthetic R^* CPU profiling in a content-addressed JSON artifact. Executable tests cover the target-conditioned dual, target-ray scaling, semantic-first target realization under arbitrary overlap, quotient clone invariance, sampler-weight recomputation, literal-duplicate rejection, exact questioner-message equality, the $7/120$ prompt-mixture shift, quotient restoration to zero, one-cell support perturbations, per-tool LP results, and the offline analysis contract. The added E4/E6 analyses use no new support annotations, task outcomes, model calls, or GPU runs. All dynamic thresholds were frozen before outcomes. The failed Qwen3-8B bank remains a transparent non-result; the qualified SkillRL bridge separately records 18/24 competence calibration, 24 paired A/B/C/D units, exact A/D coupling, and the statistical-resolution boundary that prevents upgrading a realization STOP into a population-level no-effect claim.

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